HW 8

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Perform the following activities in Python:

- Get the dataset - pick any one of the datasets from the Datasets folder.
- Load the .csv file into a data frame in Python.
- Import necessary libraries.
- Describe the data, provide information on the dataset.
- Preprocess data if needed.
- Decide what is a good split and split the data using metrics like Gini Index, Entropy and Information Gain
- Train the decision tree
- Make Predictions
- Evaluate Model Accuracy
- Visualize the Decision Tree
- Provide conclusions based on the analysis

Decision Tree Classification

Dataset: Bank Personal Loan Modelling **Target:** Personal Loan (0 = Rejected, 1 = Accepted) **Size:** 5,000 customers, 14 features

Import necessary libraries.

In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix, accuracy_score, classificatior
from sklearn.preprocessing import LabelEncoder
```

Get the dataset - pick any one of the datasets from the Datasets folder.

Dataset: Bank Personal Loan Modelling

Load the .csv file into a data frame in Python.

In [3]:
```
# Load dataset
df = pd.read_csv('/Users/mtruong/python-projects/Bank_Personal_Loan_Modellir
df.head()
```

Out[3]:

	ID	Age	Experience	Income	ZIP Code	Family	CCAvg	Education	Mortgage	Persor Lo
0	1	25	1	49	91107	4	1.6	1	0	
1	2	45	19	34	90089	3	1.5	1	0	
2	3	39	15	11	94720	1	1.0	1	0	
3	4	35	9	100	94112	1	2.7	2	0	
4	5	35	8	45	91330	4	1.0	2	0	

Describe the data, provide information on the dataset.

In [4]:
```
# Summary statistics
df.describe()
```

Out[4]:

	ID	Age	Experience	Income	ZIP Code	F
count	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000	5000.00
mean	2500.500000	45.338400	20.104600	73.774200	93152.503000	2.39
std	1443.520003	11.463166	11.467954	46.033729	2121.852197	1.14
min	1.000000	23.000000	-3.000000	8.000000	9307.000000	1.00
25%	1250.750000	35.000000	10.000000	39.000000	91911.000000	1.00
50%	2500.500000	45.000000	20.000000	64.000000	93437.000000	2.00
75%	3750.250000	55.000000	30.000000	98.000000	94608.000000	3.00
max	5000.000000	67.000000	43.000000	224.000000	96651.000000	4.00

In [5]:
```
# Check for missing values
df.isnull().sum()
```

Out[5]:
```
ID                    0
Age                   0
Experience            0
Income                0
ZIP Code              0
Family                0
CCAvg                 0
Education             0
Mortgage              0
Personal Loan         0
Securities Account    0
CD Account            0
Online                0
CreditCard            0
dtype: int64
```

In [6]:
```
# Class distribution: Personal Loan - Accepted vs. Rejected
df['Personal Loan'].value_counts()
```

Out[6]:
```
Personal Loan
0    4520
1     480
Name: count, dtype: int64
```

Preprocess data if needed.

- Drop ID and ZIP Code because they are not predictive features
- Define features and target
- Split 80/20

In [7]:
```
# Drop the columns that are not predictive features
df.drop(columns=['ID', 'ZIP Code'], inplace=True)

# Define features and target
X = df.drop(columns=['Personal Loan'])
y = df['Personal Loan']

# Split into 80% train, 20% test, and stratify to preserve class balance
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, ra
```

Decide what is a good split and split the data using metrics like Gini Index, Entropy and Information Gain

In [8]:
```
# Compare Gini Index vs Entropy (Information Gain) to determine best split c
for criterion in ['gini', 'entropy']:
    model = DecisionTreeClassifier(criterion=criterion, random_state=42)
    model.fit(X_train, y_train)
    print(f"Criterion: {criterion:<10} Accuracy: {accuracy_score(y_test, moc
```

```
Criterion: gini       Accuracy: 0.9820
Criterion: entropy    Accuracy: 0.9920
```

Train the decision tree

In [9]:
```
# Train final model using best criterion
dt = DecisionTreeClassifier(criterion='entropy', max_depth=5, random_state=4
dt.fit(X_train, y_train)
```

Out[9]:
```
▾ DecisionTreeClassifier
▶ Parameters
```

Make Predictions

In [10]:
```
# Predict on the test set
y_pred = dt.predict(X_test)
```

Evaluate Model Accuracy

In [11]:
```
# Overall accuracy
accuracy_score(y_test, y_pred)
```

Out[11]:
```
0.991
```

In [12]:
```
# Precision, recall, and F1-score per class
print(classification_report(y_test, y_pred, target_names=['Rejected', 'Accep
```

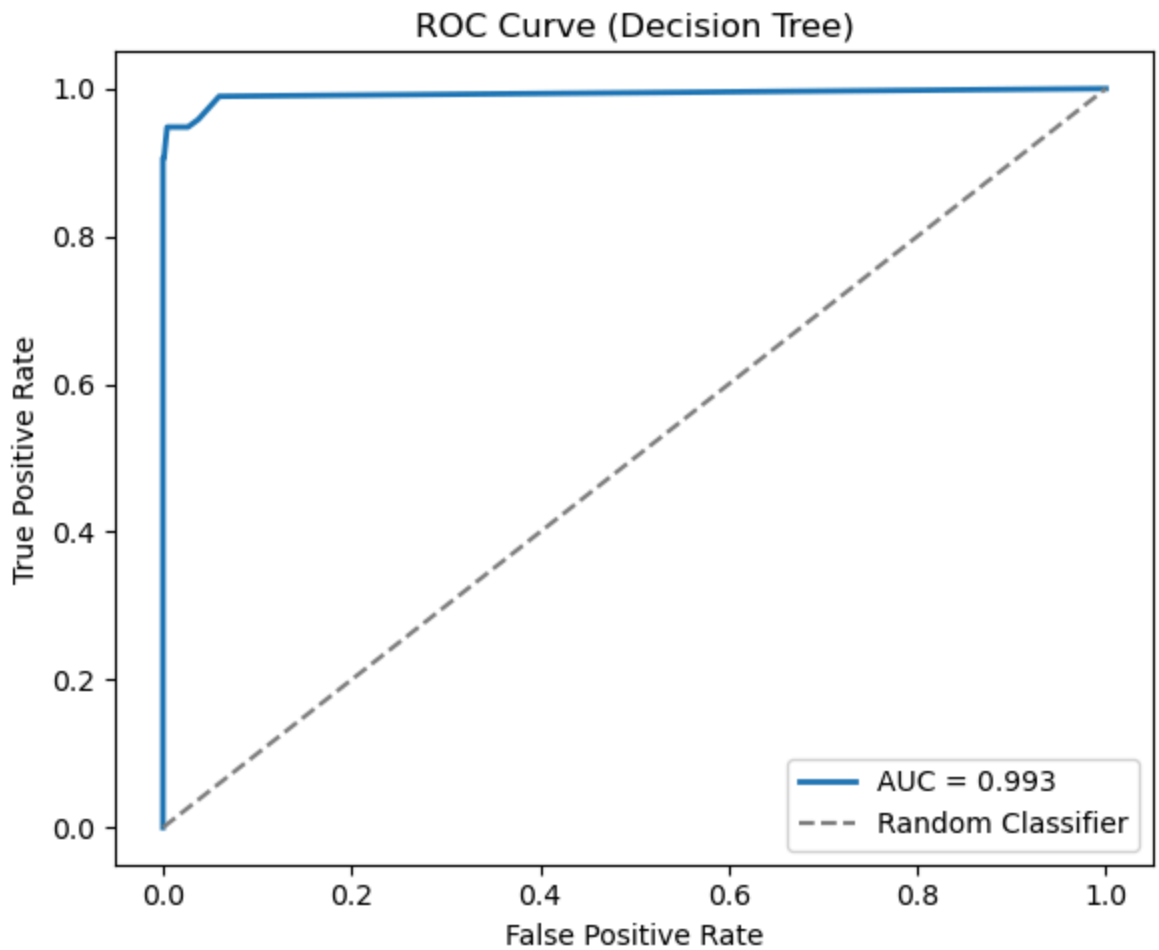
```
              precision    recall  f1-score   support

    Rejected       0.99      1.00      1.00       904
    Accepted       0.96      0.95      0.95        96

    accuracy                           0.99      1000
   macro avg       0.98      0.97      0.97      1000
weighted avg       0.99      0.99      0.99      1000
```

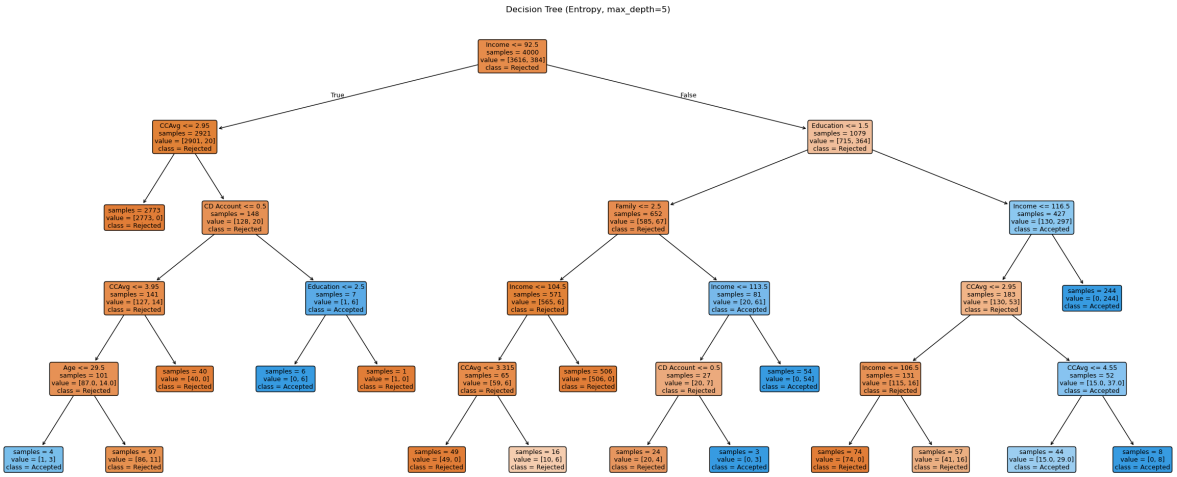
In [13]:
```
# Display ROC Curve because it measures the model's ability to separate clas
y_proba = dt.predict_proba(X_test)[:, 1]
fpr, tpr, _ = roc_curve(y_test, y_proba)
roc_auc = auc(fpr, tpr)

plt.figure(figsize=(6, 5))
plt.plot(fpr, tpr, lw=2, label=f'AUC = {roc_auc:.3f}')
plt.plot([0, 1], [0, 1], color='grey', linestyle='--', label='Random Classif
plt.title('ROC Curve (Decision Tree)')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.legend()
plt.tight_layout()
plt.show()
```



Visualize the Decision Tree

In [16]:
```
# Visualize the trained decision tree
plt.figure(figsize=(24, 10))
plot_tree(dt, feature_names=X.columns, class_names=['Rejected', 'Accepted'],
          filled=True, rounded=True, fontsize=9,
          impurity=False, proportion=False)
plt.title('Decision Tree')
plt.tight_layout()
plt.show()
```



Provide conclusions based on the analysis

Conclusion: The Decision Tree using Entropy (Information Gain) as the split criterion successfully classified personal loan acceptance with high accuracy, with Income, CCAvg, and Education appearing the most important splitting features.